

Compressed Sensing of LLMs

Query-Efficient Recovery of a Black-Box Next-Token Function

Thesis Proposal (MSc or PhD)

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1 The Abstraction

We model a language model as a single deterministic nonlinear function

$$f_\theta : \mathcal{V}^L \rightarrow \mathbb{R}^\mathcal{V}, \quad c \mapsto z = f_\theta(c),$$

mapping a context window c (a token sequence of length up to L) to the logit vector z over the next token, where $\mathcal{V}$ is the vocabulary.

Auditing a closed LLM can be framed as a function recovery problem: Given observed conversations, estimate the underlying function f_θ .

The thesis asks the compressed-sensing question in its purest form: treating f_θ as an unknown high-dimensional signal, how much of it can be recovered from $m \ll |\mathcal{V}|^L$ evaluations, under what structural conditions, and with what query designs?

There are no separate “attacks,” only different *targets* — the whole function, a restriction, or a functional of it — recovered through one evaluation operator.

2 Why Compressed Sensing Applies: the Compressibility Priors

A generic function $\mathcal{V}^L \rightarrow \mathbb{R}^\mathcal{V}$ is unrecoverable from few samples — the domain has size $|\mathcal{V}|^L$. Recovery is possible only because f_θ is *not* generic. Four structural facts play the role that sparsity plays in classical compressed sensing [1–3], and that earlier underpinned prediction-API model stealing [4]:

- (P1) Low-rank codomain.** The logits factor as $z = W_U h(c)$ with hidden state $h(c) \in \mathbb{R}^d$ and un-embedding $W_U \in \mathbb{R}^{\mathcal{V} \times d}$, $d \ll |\mathcal{V}|$ [5]. Hence $\text{range}(f_\theta)$ lies in a d -dimensional subspace, and since $\text{softmax}(z + c\mathbf{1}) = \text{softmax}(z)$ the next-token distribution identifies z only modulo the constant direction $\text{span}\{\mathbf{1}\}$: the effective output dimension is d (at most $d - 1$ for the distribution), not $|\mathcal{V}|$ — the softmax bottleneck [6].
- (P2) Continuous factorisation.** f_θ factors through token embeddings $c \mapsto E(c) \in \mathbb{R}^{L \times d}$, so it is the discrete lift of a smooth map on a continuous, low-dimensional domain rather than an arbitrary function on a combinatorial set.
- (P3) Lipschitz smoothness.** On the compact embedding domain (bounded token embeddings plus positional encodings), f_θ is a composition of C^1 maps and is therefore Lipschitz, with constant controlled by the product of per-layer operator norms. The restriction to a bounded domain is essential: standard dot-product self-attention is provably *not* Lipschitz on the unbounded domain $\mathbb{R}^{L \times d}$ [7, Thm. 3.1], but on a ball of radius R its local Lipschitz constant scales as $\sqrt{L} \cdot \text{poly}(R, \|W\|)$ [8, Thm. 3.3] (see also 9). Thus nearby contexts produce nearby logits — the condition under which point evaluations are informative (Section 4).

(P4) Small description complexity. For a bounded-weight, fixed architecture the metric entropy of the realizable class grows only polynomially in $1/\varepsilon$ — roughly $\mathcal{H}_\varepsilon(\mathcal{F}) \lesssim W \log(1/\varepsilon)$ with W the parameter count — rather than the $(1/\varepsilon)^D$ entropy of generic Lipschitz functions on $[0, 1]^D$, which is exponential in the ambient dimension D [10, 11]. For the transformer class specifically, norm- and rank-based covering-number bounds make the dependence on embedding dimension and context length only logarithmic [12, 13]; networks remain Kolmogorov–Donoho rate-distortion optimal approximants of structured function classes [14].

These four properties are the compressibility priors. The thesis makes precise how each one reduces the number of evaluations needed.

3 Measurement Model

Black-box access furnishes an *evaluation operator* $\Phi : \mathcal{F} \rightarrow (\mathbb{R}^V)^m$, $\Phi(f) = (f(c_1), \dots, f(c_m))$, instantiated at several realism levels:

- **Full-logit oracle:** $y_i = f_\theta(c_i) \in \mathbb{R}^V$ (clean linear measurements of the codomain).
- **Top- k log-probability oracle:** only the k largest entries are observed — a *compressed/partial* measurement of each output vector.
- **Sampled-token oracle:** only stochastic draws from $\text{softmax}(z)$ — *noisy, quantised* measurements requiring repetition to denoise.
- **Adaptive logit-bias oracle:** query-time perturbations (Carlini et al. [15]) act as designed measurement vectors that reconstruct individual logit coordinates through a top- k API, after which the codomain subspace is identified by SVD — subspace probing under restricted output access.

A central practical theme is how these restricted oracles degrade the recovery guarantees of the full-logit case — i.e. how API defenses act as perturbations of the measurement operator.

4 Core Theoretical Object: Restricted Isometry over $\mathcal{F}$ via Point Evaluation

Fix a context distribution $\mathcal{D}$ and the population seminorm $\|f\|^2 = \mathbb{E}_{c \sim \mathcal{D}} \|f(c)\|^2$. Draw probe contexts $c_1, \dots, c_m \sim \mathcal{D}$ and form the empirical evaluation seminorm $\widehat{\|f\|}^2 = \frac{1}{m} \sum_i \|f(c_i)\|^2$. The recovery problem is well-posed precisely when the evaluation operator is a restricted isometry over the difference set $\mathcal{F} - \mathcal{F}$.

Condition 1 (Evaluation-RIP over $\mathcal{F}$). There exists $\delta \in (0, 1)$ such that, for all $f, g \in \mathcal{F}$,

$$(1 - \delta) \|f - g\|^2 \leq \frac{1}{m} \sum_{i=1}^m \|f(c_i) - g(c_i)\|^2 \leq (1 + \delta) \|f - g\|^2.$$

Condition 1 says: any two models agreeing on the m probe contexts must be globally close. When it holds with small δ , evaluations at $\{c_i\}$ form a norm-preserving sketch of the entire function class, and recovery (find $\hat{f} \in \mathcal{F}$ consistent with the measurements) is stable. The thesis develops this object along three established lines.

Sample complexity via metric entropy. The uniform isometry is an empirical-process statement; the required m is governed by the metric entropy $\mathcal{H}_\varepsilon(\mathcal{F})$ of the transformer class. Norm- and rank-based covering-number bounds for self-attention show $\mathcal{H}_\varepsilon(\mathcal{F})$ scales polynomially in $1/\varepsilon$ and only logarithmically (or not at all) in the sequence length [12, 13], so heuristically $m \gtrsim \mathcal{H}_\varepsilon(\mathcal{F})/(1 - \delta)$. The closest prior on recovering a *system* from input–output traces in a metric-entropy-optimal manner is Hutter et al. [16].

Well-posedness via smoothness (sampling numbers). Whether point samples suffice — rather than arbitrary linear functionals — is the subject of information-based complexity. Recent results show that whenever the approximation numbers are square-summable (decay rate $s > \frac{1}{2}$), the sampling numbers decay at the same polynomial rate, so point evaluations are essentially as powerful as optimal linear measurements [17–19]; the sharp, logarithm-free form $g_{cm}^2 \lesssim \frac{1}{m} \sum_{k \geq m} a_k^2$ is due to Dolbeault et al. [20]. The Lipschitz prior (P3) is exactly what places $\mathcal{F}$ in this regime, justifying that query access loses little against an idealised measurement operator. Note that these results establish the *rate* of an optimal sampling algorithm; deriving the two-sided isometry of Condition 1 still requires an empirical-process argument over $\mathcal{F} - \mathcal{F}$.

Lower bounds via packing. Applying Fano’s inequality to a maximal ε -packing of $\mathcal{F}$ converts metric entropy into a minimax lower bound: the achievable accuracy ε_m is fixed by the entropy-balance $\mathcal{H}_{\varepsilon_m}(\mathcal{F}) \asymp m \varepsilon_m^2$ for the noisy (sampled-token) oracle [21], so no recovery scheme — adaptive or not — beats this rate. This pins query complexity to the *intrinsic* complexity of the model class, independent of vocabulary size and context length except through $\mathcal{H}_\varepsilon$.

5 Recoverable Targets

Recovering all of f_θ to high accuracy is infeasible when $\mathcal{H}_\varepsilon(\mathcal{F})$ is large; the tractable regime is recovery of low-complexity *targets* derived from f_θ , all under the single evaluation operator of Section 3.

Target 1 (Codomain structure). The subspace $\text{range}(W_U)$ and the hidden dimension d — the latter being identifiable, since W_U itself is recoverable only up to an $\mathcal{V}$ -irrelevant $d \times d$ symmetry (affine in practice, orthogonal in principle). This is a functional of f_θ , recoverable as numerical-rank / subspace recovery (SVD of stacked logit vectors) from $O(d)$ full-logit queries; under a restricted top- k /logit-bias API the cost rises to $O(d|\mathcal{V}|)$, since each full logit vector must first be reconstructed coordinate-by-coordinate. Carlini et al. [15] is the empirical existence proof and the controlled benchmark.

Target 2 (Restriction to a structured sub-domain). $f_\theta|_{\mathcal{S}}$, where $\mathcal{S}$ is a prompt family varying along $k \ll Ld$ axes (e.g. in-context demonstrations of a function class [22]). Recovery is local nonlinear function approximation with sample complexity controlled by $\mathcal{H}_\varepsilon(\mathcal{F}|_{\mathcal{S}})$, small when $\mathcal{S}$ is low-dimensional. This subsumes “identify the algorithm implemented in-context” as recovery of f_θ on a structured slice.

Target 3 (General functionals). $P(f_\theta)$ — a decision boundary on a sub-domain, the presence of a watermark, a behavioural property. Such a functional is recoverable under a restricted isometry on the relevant quotient of $\mathcal{F}$, typically with far fewer queries than full recovery.

6 Research Questions

RQ1.

When does the point-evaluation operator satisfy a restricted isometry over the transformer-realizable class $\mathcal{F}$, and how do the four compressibility priors (P1–P4) enter the isometry constant?

RQ2.

What is the query complexity m for each target in terms of the metric entropy / sampling numbers of the relevant (restricted or quotiented) class, and does it match the packing lower bound?

RQ3.

How do the restricted oracles of Section 3 — top- k truncation, sampling noise, finite precision, rate limits — degrade the isometry constant, and which defenses provably destroy recoverability versus merely inflating m ?

7 Methodology

Theory track. Establish the evaluation-RIP condition for $\mathcal{F}$; import metric-entropy bounds for transformer-type classes and sampling-number results to convert them into query-complexity upper bounds; prove matching packing lower bounds; analyse restricted oracles as perturbations of the isometry.

Empirical track. On open-weight models where f_θ is fully known (Pythia, Qwen, Llama across scales): (i) empirically estimate the evaluation-isometry constant for different probe designs (random vs structured vs adaptively chosen contexts); (ii) recover targets T1–T3 and compare achieved query counts against the theory; (iii) stress-test under simulated top- k and sampling oracles. Deliverables include a reproducible probing/recovery harness and empirical query-complexity scaling curves.

8 Evaluation

- **Recovery accuracy:** subspace/parameter error (T1), function-approximation error on $\mathcal{S}$ (T2), functional error (T3).
- **Query efficiency:** measured m vs predicted complexity; scaling against d , $|\mathcal{V}|$, and the dimensionality of $\mathcal{S}$.
- **Isometry diagnostics:** empirical δ as a function of probe design and budget.
- **Robustness:** degradation across the oracle hierarchy; lower-bound tightness (achieved vs information-theoretic m).

9 Expected Contributions

1. A single compressed-sensing formalism for black-box LLMs: the model as one nonlinear function, query access as point evaluation, recovery governed by a restricted isometry over the function class.
2. Query-complexity upper bounds (via metric entropy and sampling numbers) and matching packing lower bounds for codomain structure, sub-domain restrictions, and functionals.
3. A robustness theory relating realistic API oracles to the isometry constant, with provable separations between benign and recovery-destroying defenses.
4. An open, reproducible harness and empirical scaling laws relating query budget to recoverable structure.

The framework quantifies *how auditable* a deployed model is from the outside, and conversely which deployment choices provably limit external recovery — a direct contribution to trustworthy-AI and model-auditing goals.

10 Risks and Contingencies

- *The point-evaluation operator may resist a clean RIP for the full class.* Mitigation: smoothness places $\mathcal{F}$ in the sampling-numbers regime; where a global RIP is unavailable, restrict attention to targets T1–T3, whose quotient classes have small metric entropy and admit isometries directly.
- *Restricted oracles may block logit access entirely.* Mitigation: the open-weight track is independent of any external API; oracle restrictions are simulated locally with controlled corruption layers.
- *Theory–experiment gap.* Mitigation: T1 (validated by Carlini et al. [15]) guarantees a defensible empirical result even if the bounds for T2/T3 come out only partial.

11 Indicative Timeline (6 months, master’s-thesis scoping)

Month	Milestone
1	Formalise the function/measurement model; consolidate metric-entropy and sampling-number tools
2	Evaluation-RIP theory; codomain recovery (T1) + benchmark harness
3	T1 empirical validation; oracle-robustness study
4	Sub-domain restriction (T2) and functionals (T3): complexity upper/lower bounds
5	T2/T3 experiments; query-complexity scaling laws
6	Synthesis, writing, defense preparation

12 Prerequisites for the Student

Linear algebra and high-dimensional probability; prior exposure to compressed sensing or statistical learning theory (sample complexity, covering/packing arguments); comfort with PyTorch and running open-weight LLMs. Familiarity with metric-entropy / information-based-complexity arguments is an asset but can be acquired during Month 1.

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